



Advanced Classical Mechanics

Lecture 6: Conservation Laws & Gravity

Readings: Morin - Chapter 5, Marion Chapter 2 & 5

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Newtonian Dynamics

Problem 1. (Force components in polar coordinates):

Recall the definition of Polar Coordinates from Question 6 of our lecture 1 (ACM1). The Cartesian Coordinates can be written in terms of polar coordinates as,

$$x = r \cos \theta ; y = r \sin \theta .$$

We would like to express the equation of motion in Polar Coordinates as opposed to Cartesian Coordinates ($\mathbf{F} = m\mathbf{a} \equiv m\ddot{\mathbf{r}}$).

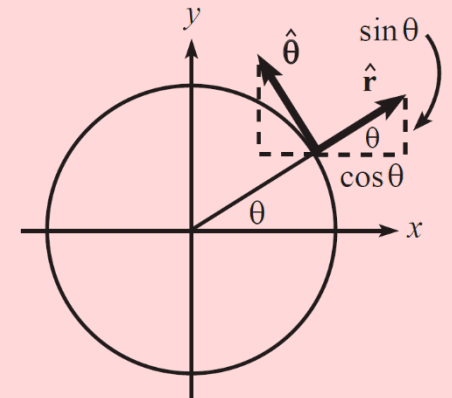
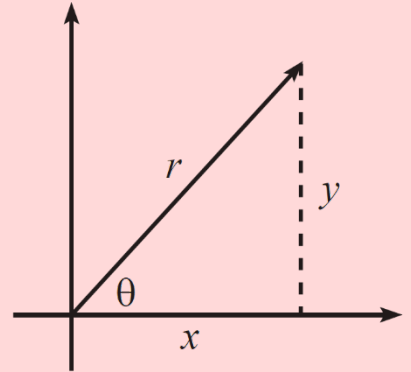
Using the representation of the Polar bases as illustrated in the figure, show that the Polar equation of motion can be written as,

$$\mathbf{F} \equiv F_r \hat{\mathbf{r}} + F_\theta \hat{\boldsymbol{\theta}} = m\ddot{\mathbf{r}},$$

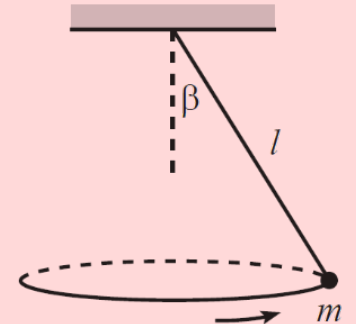
where,

$$F_r = m(\ddot{r} - r\dot{\theta}^2),$$

$$F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}).$$



Problem 2. (Circular pendulum): A mass hangs from a massless string of length l . Conditions have been set up so that the mass swings around in a horizontal circle, with the string making a constant angle β with the vertical. What is the angular frequency, ω , of this motion?



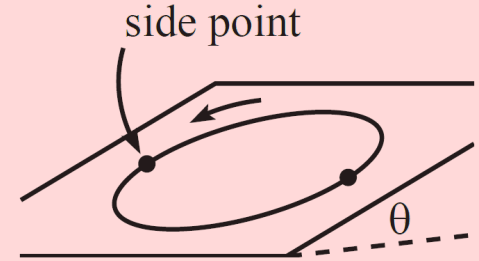
Newtonian Dynamics

Problem 3. (Driving on tilted ground):

A driver encounters a large tilted parking lot, where the angle of the ground with respect to the horizontal is θ . The driver wishes to drive in a circle of radius R at constant speed. The coefficient of friction between the tires and the ground is μ .

(a) What is the largest speed the driver can have if he wants to avoid slipping?

(b) What is the largest speed the driver can have, assuming he is concerned only with whether or not he slips at one of the “side” points on the circle (that is, halfway between the top and bottom points; see the Figure)?



Problem 4. (A force $F_\theta = 2m\dot{r}\dot{\theta}$): Consider a particle that feels an angular force only, of the form $F_\theta = 2m\dot{r}\dot{\theta}$. Show that $r = Ae^\theta + B^{-\theta}$, where A and B are constants of integration, determined by the initial conditions.

Conservation laws of Physics

- What we mean by the “conservation of a physical quantity” is the fact that its value must remain constant over time.
- Three main conservation laws of physics.
 - Conservation of energy (and mass).
 - Conservation of linear momentum.
 - Conservation of angular momentum.
- These conservation laws are not axioms (i.e., accepted without proof), rather can be derived from the Newtonian Dynamics.
- Historically, there was confusion about conservation of energy vs. momentum.
 - During 1676-1689, Gottfried Leibniz realized (using the works of the contemporary physicist Christiaan Huygens) that the quantity mv^2 appears to be constant when there is no interaction (e.g., no friction). He called this conserved quantity ***vis viva*** (meaning the ***living force***).
 - The contemporary physicist Isaac Newton contested this idea and had suggested that it is another quantity that is conserved and is of the form mv .
 - It took about half a century for other researchers to realize that both Newton and Leibniz were right and that their discoveries referred to two independent conservation laws of Physics.
- We define a force on an object as **conservative force** if and only if it depends on the location of the object (as opposed to velocity or other physical quantities).
- We define **work** for a given force as the energy transferred **to or from** an object as the object is displaced along the direction of the force,

$$w = \mathbf{F} \cdot d\mathbf{x}$$



Gottfried Leibniz



Isaac Newton

Conservation laws of Physics

Problem 5. (Derivation of Conservation of Energy from Newtonian Dynamics):

Consider a force, in just one dimension, that **depends only on position**: $F = F(x)$. If the force acts on a particle of mass m with initial location x_0 and velocity v_0 , show that the integral of the force $F(x)$ over distance traveled leads to the law of conservation of energy as we know it in the form,

$$\frac{1}{2}mv^2 - \int_{x_0}^x F(x')dx' = E - U = \text{const.}$$

The integral in the above equation has a specific name in Physics: The work done by the force $F(x)$ over the distance traveled from x_0 to x . In other words, the above expression states that **the change in the kinetic energy of a system equals the work done by the force on it during the displacement**.

Problem 6. (Gravitational potential energy): Consider two-point masses, M and m , separated by a distance r . Newton's law of gravitation says that the force between them is attractive and has magnitude GMm/r^2 . Show that the potential energy of the system with respect to a separation distance r_0 of infinity is given by,

$$U(r) = -\frac{GMm}{r}$$

Problem 7. (Gravity near the earth): Show that the gravitational potential energy of a mass m at height y , relative to the ground, is given by,

$$U(R + y) - U(R) = mgy, y \ll R$$

where g is the gravity acceleration and R is the earth radius.

Problem 8. (Unwinding string): A mass is connected to one end of a massless string, the other end of which is connected to a very thin frictionless vertical pole. The string is initially wound completely around the pole, in a very large number of tiny horizontal circles, so that the mass touches the pole. The mass is released, and the string gradually unwinds. What angle does the string make with the pole at the moment it becomes completely unwound?

Conservation laws of Physics

Problem 9. (Raising a book): Assume that you lift a book up at constant speed, so there is no change in kinetic energy. However, we know that the potential energy of the book is increasing. Explain how this is consistent with the law of conservation of energy.

Problem 10. (Conservation of energy in three dimensions) Consider the same situation as in Problem 5 but in three dimensions. Show that the Newton's law leads to the same form of the law of conservation of energy but only expressed in terms of the 3D position vector of the particle $\mathbf{r}$.

Problem 11. (Conservative force is curl-free) Recall from the previous notes that a conservative force only depends on the position within a given coordinates system and that potential energy can only be defined for conservative forces. Prove that the conservative force is equivalent to curl-free force: $\nabla \times \mathbf{F} = 0$

Problem 12. (Central force) A **central force** is a force that points radially and whose magnitude depends only on r , that is, $\mathbf{F}_C(\mathbf{r}) = F_C(r) \hat{\mathbf{r}}$. Show that **a central force is conservative** by explicitly showing that $\nabla \times \mathbf{F}_C = 0$.

Problem 13. (Gravitational force of a sphere) The gravitational force on a point mass m , located a distance r from a point mass M , is given by Newton's law of gravitation as, $F(r) = \frac{-GMm}{r^2}$, where the minus sign indicates the (attractive) direction of the force with $G = 6.67 \times 10^{-11} \frac{m^3}{kg \ s^2}$. What is the force if we replace the point mass M by a sphere (thin-shell) of radius R and uniform-density mass M ?

Problem 14. (Gravitational force of a uniform shell) Consider the same problem as in Problem 13. However, instead of a sphere, we now have uniform-density shell of inner radius $r_{\min}$ and outer radius $r_{\max}$. Derive an equation for the force of gravity felt by a hypothetical mass m at different distances from the shell (outside, inside, and within the shell)?

Conservation laws of Physics

Problem 15. (Supporting a tube) Imagine the following unrealistic undertaking. Drill a narrow tube, with cross-sectional area A from the earth's surface down to the center. Then line the cylindrical wall of the tube with a frictionless coating. Then, fill the tube with the dirt (and magma, etc.) you originally removed. What force is necessary at the bottom of the dirt tube (that is, at the center of the earth) to hold it up? Let the earth's radius be R and assume (incorrectly) a uniform mass density ρ .

Problem 16. (Leaving the sphere) A small mass rests on top of a fixed frictionless sphere. The mass is given a tiny kick and slides downward. At what point does it lose contact with the sphere?

Problem 17. (Escape velocity) (a) Find the escape velocity (that is, the velocity above which a particle escapes to $r = \infty$) for a particle on a spherical planet of radius R and mass M . What is the numerical value for the earth? The moon? The sun?

Problem 18. (Pendulum projectile) A pendulum is held with its string horizontally and is then released. The mass swings down, and then on its way back up, the string is cut when it makes an angle θ with the vertical. What should θ be so that the mass travels the largest horizontal distance by the time it returns to the height it had when the string was cut?

